

DENETH PRIYADARSHANA

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PERSONAL STATEMENT

Enthusiastic Electronic and Telecommunication Engineering undergraduate at the University of Moratuwa with strong interests in robotics, embedded systems, computer vision and control systems. Eager to design intelligent robotic systems through the integration of electronics, software and mechanical design.

EDUCATION

University of Moratuwa <i>B.Sc. (Hons) in Electronic and Telecommunication Engineering</i> • CGPA (Semesters 1–4): 3.83 / 4.0 • Dean's List: Semester 1, Semester 2, Semester 4	Mar. 2024 – Present Moratuwa, Sri Lanka
Ananda College <i>G.C.E. Advanced Level (Physical Science – English Medium)</i> • Result: 4 A's (Combined Mathematics, Physics, Chemistry, General English) • District Rank (Colombo): 19 • Island Rank: 62 • Z-Score: 2.7516	Aug. 2020 – Feb. 2023 Colombo 10, Sri Lanka
Thurstan College <i>G.C.E. Ordinary Level (English Medium)</i> • Result: 9 A's	Jan. 2014 – June 2020 Colombo 07, Sri Lanka

PROJECTS

6 DOF Robotic Arm <i>University Project</i> • Developed a computer-vision based 6-DOF robotic arm with a custom-designed 4mm sheet metal aluminium structure, precise object detection, localization and manipulation through forward and inverse kinematics, MATLAB-based kinematic modeling and trajectory simulation, and coordinated multi-axis motion control. • Hardware & Software: STM32, NEMA 24/23/17 Stepper Motors, 4mm Aluminium Sheet Metal, SOLIDWORKS, C/C++, Altium Designer, MATLAB	Feb. 2026 – Jul. 2026
Micromouse Autonomous Maze Solver <i>Team Project (PRISM)</i> • Developing an autonomous micromouse robot capable of high-speed exploration and optimal navigation of a 16×16 maze, integrating real-time maze mapping, flood-fill path planning, motion profiling and PID motor control (feedback & feedforward) tuned via MATLAB-based system modeling. • Hardware & Software: STM32, N20 DC Motors (1000 RPM), C/C++, Altium Designer, SOLIDWORKS, MATLAB	Jun. 2025 – Present
ROS 2 Vision-Based Autonomous Line Following Robot <i>Personal Project</i> • Developing a ROS 2 based autonomous mobile robot using Raspberry Pi 5 and camera-based computer vision for real-time line detection and navigation, with modular ROS 2 nodes for image processing, decision making and motion control. • Hardware & Software: ROS 2 (rclcpp), OpenCV, C/C++, Raspberry Pi 4, ESP32, Camera, N20 DC Motors	Jun. 2026 – Present
Autonomous Physical-Virtual Master-Slave Robot System	Jan. 2026 – Mar. 2026

Team Project (PRISM)

- Designed and programmed an autonomous robot for the Sri Lankan Robotics Challenge (SLRC) 2026 University Category, combining AprilTag-based maze navigation, multi-key coordinate decoding, object pick-and-place with a custom slider mechanism, and real-time REST API control of a networked simulated robot through 14 sequenced waypoints while avoiding a dynamically patrolling hostile agent.
- Hardware & Software: Raspberry Pi 4, Camera, Arduino Mega, Arduino Nano, OpenCV, C/C++

Autonomous Multi-Challenge Competition Robot

Aug. 2025 – Dec. 2025

University Project

- Designed and programmed a multifunctional autonomous robot capable of grid navigation, object manipulation, color detection, wall following, circular track navigation, arrow tracking, dotted-line following and ramp climbing through integrated sensing, control and real-time decision-making algorithms.
- Hardware & Software: Arduino Mega, JGA25 DC Motors, Servo Motors, C/C++

Fully Analog Line Follower

Aug. 2025 – Dec. 2025

University Project

- Developed a fully analog line-following robot capable of autonomous path tracking without microcontrollers, utilizing IR-based line sensing, comparator-driven control logic and analog feedback.
- Hardware & Software: Analog components (op-amps, transistors, etc.), N20 DC Motors (300 RPM), Altium Designer, SOLIDWORKS

Fully Analog DC Motor Speed Controller

Feb. 2026 – Jul. 2026

University Project

- Designed and implemented a fully analog DC motor speed controller for precise speed regulation using magnetic encoders as feedback.
- Hardware & Software: Analog components (op-amps, transistors, etc.), N20 DC Motor (750 RPM)

Smart Contact-based Flow Meter

Feb. 2025 – Jul. 2025

University Project

- Developed an IoT-enabled smart water flow meter capable of real-time flow rate and temperature monitoring, integrating embedded firmware, wireless communication and a cross-platform mobile app for live monitoring, historical data visualization, configurable flow thresholds and user notifications.
- Hardware & Software: ESP8266, Paddle Wheel Flow Sensor, OLED Display, React Native, Node.js, Express.js, SQLite, WebSockets, JWT, C/C++, JavaScript, Altium Designer, SOLIDWORKS

UniLink (beta)

Apr. 2024 – Aug. 2024

Personal Project

- Developed a social networking platform for university students and lecturers to share ideas, collaborate and communicate.
- Tools Used: JavaScript, React, Express.js, WebSockets, JWT, HTML, CSS, MySQL

Website for IntelliCon '24

Apr. 2024

AIESEC

- A website built for a hackathon (IntelliCon '24) organized by AIESEC, SLIIT, Sri Lanka.
- Tools Used: JavaScript, Node.js, Express.js, Handlebars (hbs), HTML, CSS, SQLite

SAMP Online Gaming Server

2017 – 2019

Personal Project

- Created and maintained an online multiplayer server for Grand Theft Auto: San Andreas, designing and implementing custom gameplay mechanics, player management systems, administration tools and server-side scripting for a persistent multiplayer environment using Pawn.
- Tools Used: Pawn

COMPETITIONS

Micromaze 2.0 — IIT

Aug. 2025

Finalists – Selected among Top 10 teams (Team Maze Hackers)

- Designed and developed an autonomous micromouse robot which incorporates motion control, wall detection and path planning to navigate a maze as fast as possible.

SLRC 2026 — UOM

Mar. 2026

Finalists – Selected among Top 8 teams (Team PRISM)

- Designed and developed an autonomous mobile robot incorporating computer vision, sensor fusion, motion control and autonomous navigation to complete competition tasks.

EXPERIENCE

Private Tutor — Mathematics

2023 – 2025

- Conducted individual classes for G.C.E. Advanced Level Combined Mathematics subject.

Private Tutor — Chemistry

2023

- Conducted individual classes for G.C.E. Advanced Level Chemistry subject.

CJ Institute — Invigilator, Question Designer

2023

Delkanda, Sri Lanka

- Worked part-time to address doubts/issues of students regarding G.C.E. Advanced Level Physical Science subjects.
- Designed new model questions and model papers for G.C.E. Advanced Level Combined Mathematics.

AWARDS & HONORS

Mahapola Merit Scholarship

2025

- Government scholarship for merit results (ranked 62nd in Sri Lanka at G.C.E. A/L).

Nenaharasara Award

2023

- Awarded by Ananda College for distinctions in all subjects at G.C.E. A/L.

EXTRACURRICULAR

Swimming — Thurstan College

2017 – 2018

Basketball — Thurstan College

2015 – 2016

TECHNOLOGIES

Languages: C, C++, Python, JavaScript, SQL, HTML, CSS, Pawn

Frameworks: ROS 2, React.js, React Native, Express.js

Libraries: OpenCV, rclcpp

Databases: MySQL, SQLite, MongoDB

Software: Altium Designer, SOLIDWORKS, MATLAB, LTspice

SKILLS & INTERESTS

Skills: Analytical thinking, Problem solving, Adaptability, Teamwork

Interests: Robotics, Embedded Systems, Computer Vision, Control Systems, Electronics, Autonomous Systems

REFERENCES

Dr. Upeka Premaratne

Senior Lecturer – Grade I, Dept. of Electronic and Telecommunication Engineering, University of Moratuwa

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